Vincent X. Wang

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Research interests: (non-convex) optimization theory for machine learning, optimization with privacy constraints, geometry, geometry of optimization

Education

University of Chicago — Chicago, IL

B.S. in Mathematics and Computer Science — *Expected June 27*

- Transferred from Rice University (Aug 2023 – May 2025)

Publications ($(\alpha\beta)$ denotes alphabetical ordering)

1. V. X. Wang, V. K. Garg. *Mirror Descent at the Edge of Stability.* Accepted at **OPT2026** Workshop at NeurIPS; submitted to ICLR 2027 (manuscript available upon request).
2. $(\alpha\beta)$ C. Chen, T. Gabrielsen, R. Morrison, N. Pasman, M. Reeve, V. X. Wang. *Graph Gonality Under Uniform Subdivision.* In preparation (manuscript available upon request).
3. $(\alpha\beta)$ D. Leitz, R. Morrison, S. Newman-Taylor, V. X. Wang. *The d-gonal Locus in the Moduli Space of Tropical Plane Curves.* Submitted to *Algebraic Combinatorics*, arXiv.
4. $(\alpha\beta)$ M. Aliakbarpour, Z. Shi, R. Stevens, V. X. Wang. *Nearly-Linear Time Private Hypothesis Selection with the Optimal Approximation Factor.* 39th Conference on Neural Information Processing Systems, **NeurIPS 2025**, arXiv.

Research Experience

Analysis of gradient methods — June 2026 – October 2026

Aalto University

- Theoretical and empirical analysis of gradient descent on non-convex losses in the edge of stability regime (publication at OPT 2026)
- Advisor: Vikas Garg

Differentially private stochastic convex optimization — Jan 2026 – March 2026

Toyota Technological Institute – Chicago

- Theoretical analysis of differentially private stochastic convex optimization (DP-SCO) minimax rates in ℓ_p norm
- A new privacy analysis technique for zCDP via log-concave isoperimetry inequalities
- Advisor: Mahdi Haghifam

SMALL REU: Tropical geometry and chip-firing games — June 2025 – Aug 2025

Williams College, Department of Mathematics

- Conducted theoretical research in algebraic combinatorics and graph gonality (two preprints)
- Created open-source high-performance Julia library for analysis of chip-firing graphs
- Advisor: Ralph Morrison

Differentially private statistical machine learning algorithms — Jan 2025 – June 2025

Rice University, Department of Computer Science

- Developed differentially private algorithms, such as interior point and histogram estimation, assuming access to an unverified prediction
- Proposed a nearly-linear time private hypothesis selection algorithm (publication at NeurIPS 2025)
- Advisor: Maryam Aliakbarpour

Discontinuous Galerkin methods for wave equations (GitHub) — Apr 2024 – Dec 2024

Rice University, Department of Computational Applied Mathematics and Operations Research

- Formulated new matrix algorithms for Bernstein basis operators in high-fidelity PDE simulations, resulting in over 100x speedups at high orders ($N > 8$) over state-of-the-art
- Awarded Outstanding Presentation in Computational Applied Mathematics and Operations Research at Gulf Coast Undergraduate Research Symposium
- Advisor: Jesse Chan

Awards

- 2026 **Barry Goldwater Scholar** — Mar 2026
- JMM 2026 Undergraduate Travel Grant — Nov 2025
- **Putnam** Top 500 (Score: 30) — Dec 2025
- US Physics Olympiad Silver Medal (2x Qualifier) — May 2022
- 3x AIME Qualifier — 2021-2023
- USA Coding Olympiad Silver Contestant — Jan 2020
- Lam Research Core Values Scholarship — May 2023

Presentations

- Aalto Science Institute, *Espoo, Finland.* — Aug 2026
  Poster: Mirror Descent at the Edge of Stability
- JMM 2026 (AMS Contributed Paper Session on Combinatorics, IV), *Washington, D.C.* — Jan 2026
  Talk: Minimality Results on Graphs with Different Discrete and Metric Gonality
- JMM 2026 (PME Contributed Session on Research by Undergraduates, V), *Washington, D.C.* — Jan 2026
  Talk: Computation on Gonality of Uniformly Subdivided Graphs
- Williams College Mathematics Colloquium, *Williamstown, MA* — Aug 2025
  Talk: The d-gonal Locus in the Moduli Space of Tropical Plane Curves
- Williams College Summer Science Research **Poster** Session, *Williamstown, MA* — Aug 2025
  Poster: The d-gonal Locus in the Moduli Space of Tropical Plane Curves
- Gulf Coast Undergraduate Research Symposium, *Houston, TX* — Nov 2024
  Talk: Efficient Julia Implementations of Bernstein Basis Discontinuous Galerkin Methods
  Award: Outstanding Presentation in Computational Applied Mathematics and Operations Research
- SIAM TX-LA, *Waco, TX* — Oct 2024
  Poster: Efficient Julia Implementations of Bernstein Basis Discontinuous Galerkin Methods
- RTG Numerical Mathematics & Scientific Computing Annual Workshop, *Houston, TX* — Oct 2024
  Poster: Efficient Julia Implementations of Bernstein Basis Discontinuous Galerkin Methods
- Materials Research Society Fall Meeting, *Boston, MA* — Nov 2022
  Talk: Molecular Dynamics (MD) Simulations of Soil-Strengthening Nanocomposite-Polyelectrolyte Hydrogels

Teaching and service

- Grader: Honors Multivariable Calculus (Rice MATH232; Sp'24, Sp'25), Basic Algebra I (UChicago MATH25400, Aut'26)
- TA: Algorithmic Thinking (Rice COMP182; Sp'25)
- Reviewer: ICLR 2027
- Lovett College academic fellow (Fall'24, Sp'25)
- Goldwater Ambassador Program mentor ('26)
- AWM Rice Integration Bee problem setter and organizer

Summer schools

- University of Minnesota Summer School: Mathematics of Data Science (2026)
- Texas A&M University Undergraduate Summer School: Modeling and Simulation with PDEs (2025)

(Pre-AI) Projects

nestalgia_rs
Nintendo NES emulator in Rust from scratch (can play Donkey Kong + more!)

rchess
Rust-based chess library and engine

Some interests

Electronic music production, classical (piano) composition, topology/geometry, Teamfight Tactics, rhythm games, matcha latte making